

Real-time dispatch management of shared autonomous vehicles with on-demand and pre-booked requests

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ABSTRACT

Autonomous vehicle technology is poised to revolutionize shared vehicle systems, offering the potential for increased efficiency and convenience. To better devise management strategies for shared autonomous vehicles, this paper addresses a real-time dispatch problem with hybrid requests, where on-demand (immediate) and pre-booked (reserved) trip requests coexist. The coexistence of these two types of request behaviors introduces considerable complexity to real-time dispatch due to the uncertainty in trip demand. We design an approximate dynamic programming (ADP) approach for making vehicle–trip assignments and vehicle relocation decisions. We first formulate the real-time vehicle dispatch problem as a dynamic program and decompose it into time-staged subproblems. To effectively handle the high-dimensional state space, we replace the value functions with tractable approximations and propose a piecewise-linear functional approximation method that captures the spatiotemporal value of vehicles. To calibrate the parameters in the approximations, we propose DualT and DualNext algorithms to provide precise dual information, thereby enhancing the accuracy of our approach. Furthermore, we propose a lookahead strategy that incorporates pre-booked request information into the ADP approach for improving real-time decision-making. We validate the effectiveness of the ADP approach through numerical experiments conducted using taxi data from Brooklyn, New York. The ADP approach outperforms benchmark policies in solution quality while maintaining computational efficiency, and the incorporation of the lookahead strategy significantly enhances the performance of the ADP approach, yielding substantial improvements. Numerical results demonstrate that integrating pre-booked requests into vehicle dispatch management can greatly enhance the system efficiency.

1. Introduction

Shared mobility provides users on-demand access to various transportation modes, such as vehicles, bicycles, or others, on an as-needed basis (Shaheen and Cohen, 2021). Notably, carsharing, ridesourcing (also known as ride-hailing), and ridesharing, have experienced substantial growth in recent years (Marsden, 2022). The advent of autonomous vehicles (AVs), a technological evolution for urban mobility, allows vehicles to sense the surroundings and navigate safely without human intervention. AVs offer a multitude of advantages compared to conventional human-driven vehicles, including enhanced safety, improved fuel efficiency, alleviation of traffic congestion, and decreased vehicle emissions. Additionally, AVs expand accessibility to mobility for individuals who are constrained by age or disabilities, fostering inclusivity in transportation (Milakis et al., 2017).

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The ongoing development of shared mobility and vehicle automation has fostered the emergence of shared autonomous vehicles (SAVs). Prominent autonomous driving technology firms, such as Waymo and Apollo, have been actively conducting trials to integrate autonomous driving technology into mobility-on-demand platforms. The integration of SAV systems holds great promise for both users and operators, thanks to AVs' inherent self-driving capabilities. SAVs can offer convenient on-demand ride services, eliminating the need for users to pick up and return shared vehicles. Furthermore, as highlighted in Litman (2020), operational costs associated with SAV will be notably lower than those incurred by human-driven taxis and ride-hailing services.

A SAV system comprises three key components: user trips, a fleet of SAVs, and an operator responsible for managing the SAV fleet. The inherent cooperative features of SAVs allow the operator to centrally make vehicle dispatch decisions so that optimal system efficiency can be attained. As studied in Guo et al. (2022), vehicle dispatch includes two critical aspects, i.e., vehicle–trip assignment and empty vehicle relocation. When the operator receives a user's trip request, the vehicle–trip assignment involves matching an available SAV with the request, followed by a prompt pick-up and transportation of the user to the destination. This decision requires the selection of a vehicle from the vicinity of the user's original location to ensure a reasonably short waiting time. Concurrently, vehicle relocation is a proactive strategy aimed at repositioning empty vehicles to designated locations, which can effectively manage vehicle inventory distribution across different locations during operational hours. The self-driving capabilities of SAVs enable this relocation process to occur at any time for any idle vehicle, without limitations imposed by human labor. The primary goal of vehicle dispatch is to optimally allocate the vehicle resources, thereby relieving the spatial–temporal discrepancy between user trip demand and vehicle supply (Mao et al., 2020). Efficient vehicle dispatch management holds the potential to significantly increase the utilization of vehicles, reduce the waiting time of users, and improve operational profitability.

A significant challenge in real-time vehicle dispatch arises from the inherent uncertainty in user trips. Although historical data collected from mobility-on-demand platforms offer insight into a statistical spatiotemporal demand trend, specific information regarding future trip requests remains unknown. Consequently, it is challenging for the SAV operator to make high-quality decisions on vehicle dispatch, as they need to anticipate future trips effectively. This challenge is amplified by the necessity to make real-time decisions immediately upon receiving trip requests.

Besides the on-demand services, mobility-on-demand platforms usually offer reservation options, enabling trip behavior to book ride services in advance. When a user sends a pre-booked trip request with a specified departure time, the operator faces the challenge of quickly deciding whether to accept it. In pursuit of profit maximization, the operator should select the most profitable trips, especially when vehicle resource is limited. However, evaluating the profitability of a trip beforehand is difficult due to the dynamic nature of vehicle supply and the uncertainty in trip demand. Furthermore, accepting pre-booked requests may affect the flexibility of vehicle dispatch for the on-demand requests, as vehicles must be available at the users' origin at the specified departure time to accommodate the pre-booked requests. Consequently, the interplay between the on-demand and pre-booked requests requires careful consideration when making vehicle dispatch decisions (Duan et al., 2020; Yu et al., 2023). The inclusion of pre-booked requests in addition to on-demand requests significantly increases the complexity of real-time vehicle dispatch.

This study focuses on a real-time vehicle dispatch problem for a SAV fleet, where both on-demand (immediate) and pre-booked (reserved) trip requests coexist. The objective is to develop effective solutions for vehicle–trip assignments and empty vehicle relocation over time. This study aims to tackle the challenges arising from trip request uncertainties and the complexity of the two types of request behaviors. The contributions can be summarized as follows.

- We formulate the real-time vehicle dispatch problem as a dynamic program, and decompose it into time-staged stochastic optimization subproblems. To model the SAV dynamics, we develop a time–space network that explicitly represents both the on-demand and pre-booked requests by constructing two types of service arcs. Our model tracks the SAV dynamics from the current state to the future, ensuring vehicle availability for serving accepted pre-booked requests.
- We propose a customized approximate dynamic programming (ADP) approach to handle the challenge posed by the high-dimensional state space. We design a piecewise-linear functional approximation to capture the spatiotemporal value of vehicles. To update the value function approximations, we develop two new algorithms, DualT and DualNext, which provide precise dual information to enhance the ADP approach. Furthermore, we propose a lookahead strategy that is integrated into the ADP approach, leveraging pre-booked request information to enhance current decision-making.
- To assess the effectiveness of the ADP approach, we develop four benchmark policies based on taxi data from Brooklyn, New York. Numerical results demonstrate our customized ADP approach offers high-quality solutions for vehicle dispatch while maintaining computational efficiency. We also confirm that the incorporation of the lookahead strategy significantly improves the performance of the ADP approach.

The remainder of this paper is organized as follows. Section 2 reviews existing research on the fleet operations of SAVs. Section 3 presents the real-time SAV dispatch problem and formulates the mathematical model. Section 4 proposes an ADP approach with DualT and DualNext algorithms. Section 5 conducts numerical experiments to evaluate the ADP approach against the benchmark policies. Finally, Section 6 gives a summary of the main achievements and suggestions for future study.

2. Related studies

In recent years, SAV services have been substantially investigated from different aspects (Narayanan et al., 2020; Zhao and Malikopoulos, 2020). A stream of studies investigated the planning problem of SAV systems, which mainly focuses on fleet sizing (Ma et al., 2017; Santos and Correia, 2021) and parking infrastructure planning (Li et al., 2021a; Seo and Asakura, 2022). Charging

infrastructure planning is also extensively investigated for SAV systems with electric vehicles (Vosooghi et al., 2020; Zhao et al., 2021; Chen and Liu, 2023).

We emphasize the related studies on the fleet management of the SAV systems. Simulation techniques are widely used for SAV modeling, and to investigate different vehicle dispatch strategies. Fagnant and Kockelman (2014) developed an agent-based model to evaluate the impact of SAVs. Four different SAV relocation strategies are tested. The results indicated that the large-block relocation strategy based on the block balance value is the most effective. Azevedo et al. (2016) integrated autonomous mobility on demand into SimMobility (an agent-based demand and supply model) to explore the effects of fleet size and station location for SAV systems. Two vehicle–trip assignments (including a greedy assignment and a minimum weight bipartite matching) and a minimal-cost relocation based on predicted demand are applied to optimize the system. Hyland and Mahmassani (2018) utilized an agent-based simulation tool to solve a dynamic fleet management problem. Six vehicle–trip assignment strategies are compared for a stochastic on-demand mobility service. Hörl et al. (2019) evaluated the performance of four SAV fleet operational strategies, including two assignment and two relocation strategies, in an agent-based simulation environment. Chen et al. (2016) examined the operations of electric SAVs under various vehicle ranges and charging infrastructure scenarios with a discrete-time agent-based model. (Dandl and Bogenberger, 2019) introduced two bipartite optimization problems for vehicle–user assignment and vehicle relocation to autonomous electric taxi simulation. Loeb and Kockelman (2019) conducted an agent-based simulation of electric SAVs, which indicates an electric SAV can replace 6 to 10 private conventional human-driven vehicles. Hu and Dong (2022) develop a simulation framework for electric autonomous taxis operations, and combined mathematical programming and artificial neural network to solve a real-time dispatch problem for electric autonomous taxis.

In contrast to simulation, research efforts have been paid to the operation optimization of SAV systems (Ge et al., 2021). Various approaches have been proposed to tackle the real-time (dynamic) vehicle dispatch problem. Kang and Levin (2021) formulated the SAV dispatch problem as a Markov decision process and developed a max-pressure dispatch policy that maximizes the passenger throughput given a fixed fleet size. Li et al. (2021b) modeled the dispatch problem as a stochastic queuing network and developed an online minimum drift plus penalty approach for a SAV system with electric vehicles. Levin (2022) further proposed a general characterization of the SAV system with extensions of ridesharing and integration with public transit. Liang et al. (2020) solved the dial-a-ride problem with ridesharing to optimize the pick-up and drop-off activities and routes of automated taxis. A rolling horizon framework with an integer non-linear programming model is developed to deal with the real-time request. Dandl et al. (2019) proposed an offline demand aggregation method to support an online joint AV–user assignment and repositioning optimization strategy. The effect of demand aggregation level is evaluated in an agent-based simulator, which indicates that fleet performance improves as demand and subregions become more spatially disaggregated. Guo et al. (2021) developed a model to solve the dynamic driver-customer matching and vehicle rebalancing problems for mobility-on-demand systems. Robust optimization was introduced to consider the future demand uncertainty.

ADP is widely adopted in the dynamic vehicle dispatch problem under uncertainty. Warrington and Ruchti (2019) proposed a two-stage stochastic program to deal with the dynamic rebalancing of shared mobility systems. The second stage is formulated as a value function representing the expected future cost and then approximated by a separable function. Yu and Shen (2019) solved a dynamic ride-pooling problem for mobility-on-demand systems based on ADP, and two heuristic decomposition schemes based on time and location are incorporated into the ADP. Al-Kanj et al. (2020) proposed an ADP approach for an autonomous ride-hailing system with electric vehicles. When combined with learning techniques, ADP is also referred to as reinforcement learning in the computer science community. Mao et al. (2020) proposed a model-free deep reinforcement learning approach to solve the autonomous taxi dispatch problem. An actor-critic algorithm with feed-forward neural networks was developed for the approximations of both policy and value functions. Tang et al. (2020) proposed an advisor–student reinforcement learning framework to solve the online vehicle dispatch problem for an electric SAV fleet. Shi et al. (2020) developed a reinforcement learning algorithm to solve the electric vehicle fleet dispatch problem. A framework with decentralized learning and centralized decision-making was proposed to minimize customer waiting time, electricity cost, and operational costs of vehicles. (Kullman et al., 2021) employed deep reinforcement learning to solve a dynamic fleet management problem in a ride-hailing system with electric vehicles. Q-value function approximations are learned by deep neural networks. Liang et al. (2022) integrated reinforcement learning and centralized programming for real-time taxi dispatch. Xie et al. (2023) developed a two-sided multiagent deep reinforcement learning approach to optimize the dynamic fleet management problem in a mobility-on-demand system with mixed autonomy.

In previous studies, various approaches have been proposed for addressing the real-time dispatch of SAVs, yet primarily focusing on only on-demand requests. The vehicle dispatch problem with hybrid on-demand and pre-booked requests has received limited attention. To the best of our knowledge, Duan et al. (2020) proposed a centralized and decentralized dispatching strategy for autonomous taxis in the hybrid request mode. An integer linear program combining a rule-based approach is proposed. Yu et al. (2023) extended the study to the field of electric SAVs. A new integer linear program is developed to consider charging decisions. However, these approaches are based on the rolling horizon framework, wherein, at each time step, predicted future requests directly using historical data are taken as input in the integer linear programs for making relocation decisions. In this study, our goal is to thoroughly consider the stochastic nature of future requests. We mathematically formulate the real-time vehicle dispatch problem as time-staged stochastic optimization subproblems. An ADP approach is proposed to solve the stochastic subproblems by using a piecewise-linear value functional approximation method.

3. Problem description and modeling

This section presents a formal description of the real-time SAV dispatch problem with uncertainty in users' trip demand.

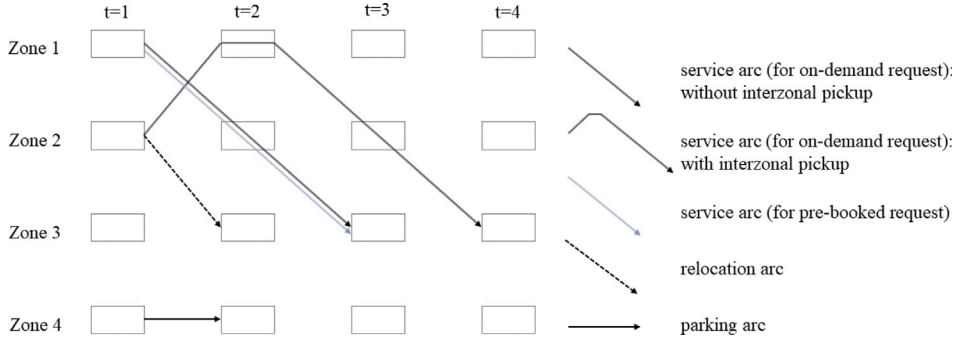


Fig. 1. An illustration of the time-space network.

3.1. Problem description

Our focus is on a SAV system without ridesharing. A given number of SAVs are operated in a one-way free-floating mode to serve user trips. The service area is partitioned into small disjoint zones, indexed by $\mathcal{Z} = \{i, j, s, \dots\}$. Each zone includes uncapped parking spots, as well as designated pick-up and drop-off locations. Consistent with the approach in Mao et al. (2020), a zone is represented by a centroid. Let t_{ij} denote the interzonal travel time of vehicles from zone i to j , which can be approximated by the travel time between the centroids of the two zones.

In the SAV system, the operational horizon is divided into identical time intervals by a set of time points, denoted as $\mathcal{T} = \{0, 1, 2, \dots, T\}$. A user sends a trip request within a time interval. The system operator makes a response at the end time of the interval. The delayed response allows the operator to consider multiple requests together. The request time is thus considered to be the end time of the interval, denoted as τ . A trip request includes information about the pick-up location zone i , the drop-off location zone j , and the specified departure time t . If the request is an on-demand request, the specified departure time t is set to be equal to τ , indicating the user's desire for an immediate pick-up and departure. Alternatively, the request can be a pre-booked request where the user provides a future departure time $t > \tau$, indicating their desired departure time for the trip. The SAV system always requires that the request time falls half or one hour before the desired time.

Upon receiving a user's trip request, the operator needs to make decisions promptly regarding the acceptance or rejection of the request. If an on-demand request is accepted, a vehicle is immediately assigned to pick up the user and transport the user to the specified destination. The assignment decision can be executed by dispatching an idle vehicle either from the same zone as the user's origin, i.e., zone i , or from a neighboring zone s . Neighboring zones should satisfy the condition that the interzonal travel time from zone s to zone i , i.e., t_{si} , cannot exceed the maximum waiting time of users W , thus ensuring $t_{si} \leq W$. If a pre-booked request is accepted, the assignment decision can be executed by dispatching a vehicle that can reach the pick-up location no later than the user's specified departure time. To enhance profitability, we introduce a dynamic vehicle swapping strategy for pre-booked requests. We can swap the vehicle-trip assignment for any other available vehicles as long as the anticipated profit would be improved, conditional on the accepted pre-booked requests still being well fulfilled without violating the desired departure times of the users. In this study, we do not consider ridesharing and assume that a vehicle can only serve one request at a time.

Besides the assignment decisions, the operator should also anticipate future trip demand and make vehicle relocation decisions. To accommodate the future spatial distribution of trip demand and increase the overall profit, vehicles can be relocated from one zone i to another zone j at time $t \in \mathcal{T}$. The proactive relocation decisions reposition vehicles to the zones with high anticipated demand. If a vehicle is not assigned with a trip request or ordered to relocate to another zone, it is assumed to be directed to park at its current location.

In our real-time vehicle dispatch problem, the request information is not revealed until the users send their requests. In the studied dynamic and stochastic environment, the operator needs to adaptively make real-time sequential decisions over time to maximize the operational horizon's overall profit.

3.2. Time-space network

To solve the aforementioned problem, we first develop a time-space network $\mathcal{G} = (\mathcal{N}, \mathcal{E})$ to model the states and actions (i.e., dispatch decisions) of the SAV system. Here, $\mathcal{N}$ denotes the set of time-space nodes, and $\mathcal{E}$ denotes the set of time-space arcs. As depicted in Fig. 1, a time-space node $n \in \mathcal{N}$ is a two-tuple $n = (i, t)$, where i indicates the zone and t represents the time. A time-space arc $e \in \mathcal{E}$ connects two nodes within the network. It is represented as $e = (i, t, j, t')$, where (i, t) and (j, t') are the nodes connected by the arc e . The start and end nodes of an arc e are denoted as n_e^+ and n_e^- , respectively. $\mathcal{E}_{i,t}^+$ and $\mathcal{E}_{i,t}^-$ denote the arcs flowing into and out of node (i, t) , respectively.

The arcs in the time-space network can be categorized into three sets of arcs, i.e., service arcs, relocation arcs, and parking arcs. Each set of arcs represents a specific type of vehicle movement within the SAV system.

- Set of service arcs $\mathcal{E}^S$: Service arcs represent vehicles' movement to serve users' trips. The on-demand and pre-booked requests have different requirements. For a trip departing from zone i to zone j at time t , the on-demand requests can be assigned to available vehicles in the zone i or a neighboring zone $s \in \mathcal{Z}(i)$. Conversely, pre-booked requests require an available vehicle arriving at zone i no later than the specified departure time t , ensuring the user departs precisely at time t . Therefore, service arcs are divided into two subsets to represent the movement that serves on-demand and pre-booked requests:
 - For vehicles serving on-demand requests from zone i to zone j departing at time t , we generate a subset of service arcs denoted as $\mathcal{E}_{i,j,t}^{S+}$. Available vehicles in the same zone i as the user's origin or a neighboring zone $s \in \mathcal{Z}(i)$ can be dispatched to pick up the user in zone i and then transport the user to the destination zone j . If a vehicle in zone i is dispatched, an arc $(i, t, j, t + t_{ij})^{S+}$ connecting two nodes (i, t) and $(j, t + t_{ij})$ is constructed, representing departure at time t from the pick-up location and arrival at the drop-off location at time $t + t_{ij}$. If a vehicle in neighboring zone s is dispatched, an arc $(s, t, j, t + t_{si} + t_{ij})^{S+}$ connecting three nodes in the sequence of (s, t) , $(i, t + t_{si})$, and $(j, t + t_{si} + t_{ij})$, is constructed, where $t + t_{si}$ is the user's departure time at the pick-up location and $t + t_{si} + t_{ij}$ is the arrival time at the drop-off location.
 - For vehicles serving pre-booked requests from zone i to zone j departing at time t , we generate a subset of service arcs denoted as $\mathcal{E}_{i,j,t}^S$. Different from the case for on-demand requests, only vehicles available in zone i at time t can be dispatched to serve pre-booked requests since the user must depart no later than time t . An arc $(i, t, j, t + t_{ij})^S$ connecting two nodes (i, t) and $(j, t + t_{ij})$ is constructed, representing an arrival time at the drop-off location of $t + t_{ij}$.
- Set of relocation arcs $\mathcal{E}^R$: Relocation arcs represent the movement of vehicles being relocated to another zones. For any time $t \in \mathcal{T}$ and zones $i, j \in \mathcal{Z}$, a relocation arc $(i, t, j, t + t_{ij})^R$ is generated to connect the nodes (i, t) and $(j, t + t_{ij})$, indicating that vehicles' relocation movement from zone i to j at time t .
- Set of parking arcs $\mathcal{E}^P$: Parking arcs represent vehicles being idle in their location zone. For any time $t \in \mathcal{T}$ and zone $i \in \mathcal{Z}$, a parking arc $(i, t, i, t + 1)^P$ is generated to connect the nodes (i, t) and $(i, t + 1)$, indicating that vehicles remain idle in their current location zone from time t to $t + 1$.

Based on the time–space network, the real-time vehicle dispatch problem involves making strategic decisions at each time point to determine whether to accept or reject trip requests, assign vehicles for accepted requests, relocate vehicles to other zones, or keep vehicles idle in their current zones. The sequential optimization process considers the dynamic and stochastic requests over time. The objective is to maximize the total net profit over the operational time horizon.

3.3. Model formulation

In this subsection, the real-time vehicle dispatch problem for the SAV system is modeled using the dynamic program framework based on the time–space network. It consists of five components: state variables, decision variables, exogenous information, the transition function, and the reward function.

3.3.1. The state variables

The state variables represent the current state of vehicles and trip requests. The variables capture information about the location and availability of vehicles, as well as the details of the trip requests.

The location and availability of each vehicle are represented by a node (i, t) , where i is the location of the vehicle and t denotes the time when the vehicle becomes available. At time τ , if a vehicle is idle, its current location at time $t = \tau$ is considered; if a vehicle is en route for service or relocation, its location at a future time $t > \tau$, i.e., the arrival time at the destination, is considered. At time τ , the number of vehicles available at time t in zone i is denoted as $R_{\tau,i,t}$.

A trip request is described by (i, j, t) , where i is the origin zone, j is the destination zone, and t is the departure time. The number of accepted pre-booked requests before time τ that should be served at time t from zone i to zone j is denoted as $D_{\tau,i,j,t}$. The number of trip requests that are newly received at time τ , departing at time t from zone i to zone j , is denoted as $\hat{D}_{\tau,i,j,t}$. These new requests can include both on-demand and pre-booked requests.

The vehicle state vector at time τ is denoted as $R_\tau = (R_{\tau,i,t})_{(i,t) \in \mathcal{N}}$. And the request state vector at time τ is denoted as $D_\tau = (D_{\tau,i,j,t})_{(i,j,t) \in \mathcal{K}}$ and $\hat{D}_\tau = (\hat{D}_{\tau,i,j,t})_{(i,j,t) \in \mathcal{K}}$, where $\mathcal{K}$ is the set of trip requests. The state vector at time τ , denoted as S_τ , is a combination of the vehicle state vector R_τ and the request state vectors D_τ and $\hat{D}_\tau$. It can be represented as $S_\tau = (R_\tau, D_\tau, \hat{D}_\tau)$. For clarification, the time indexes τ and t are used, where τ denotes the current time, while t denotes the available time for vehicles or the desired departure time for trip requests, and ensuring that have $t \geq \tau$.

3.3.2. The decision variable

The SAV operator should make responses to the trip requests at each time point. The operator determines whether to accept or reject the new requests, including both on-demand and pre-booked requests. Let $y_{\tau,i,j,t}$ represent the number of the newly accepted requests (i, j, t) at time τ . Constraints (1) must be satisfied.

$$y_{\tau,i,j,t} \leq \hat{D}_{\tau,i,j,t}, \forall (i, j, t) \in \mathcal{K} \quad (1)$$

The operator also makes dispatch decisions on vehicles. The dispatch decisions can be described by the arcs in the time-space network. The flow variable $x_{\tau,e}$ indicates the number of vehicles moving along arc e determined at time τ . The decision variables have to satisfy Constraints (2d)–(2e).

$$\sum_{e \in \mathcal{E}_{i,j,\tau}^{S+}} x_{\tau,e} - y_{\tau,i,j,\tau} = 0, \quad \forall (i, j, \tau) \in \mathcal{K} \quad (2a)$$

$$\sum_{e \in \mathcal{E}_{i,j,\tau}^S} x_{\tau,e} = D_{\tau,i,j,\tau}, \quad \forall (i, j, \tau) \in \mathcal{K} \quad (2b)$$

$$\sum_{e \in \mathcal{E}_{i,j,t}^S} x_{\tau,e} - y_{\tau,i,j,t} = D_{\tau,i,j,t}, \quad \forall (i, j, t) \in \mathcal{K}, t > \tau \quad (2c)$$

$$\sum_{e \in \mathcal{E}_{i,\tau}^-} x_{\tau,e} = R_{\tau,i,\tau}, \quad \forall (i, \tau) \in \mathcal{N}_\tau \quad (2d)$$

$$\sum_{e \in \mathcal{E}_{i,t}^-} x_{\tau,e} - \sum_{e \in \mathcal{E}_{i,t}^+} x_{\tau,e} \leq R_{\tau,i,t}, \quad \forall (i, t) \in \mathcal{N}, t > \tau \quad (2e)$$

Constraint (2a) indicates the assignment of vehicles to newly accepted on-demand requests. Constraint (2b) indicates the assignment of vehicles to accepted pre-booked requests departing at the current time τ . Constraint (2c) guarantees that the accepted pre-booked requests (i, j, t) departing at the future time t can be served by available vehicles at node (i, t) . Constraint (2d) is the flow conservation constraint to ensure that all the vehicles available at the current time τ are dispatched. Constraint (2e) indicates the number of vehicles leaving from zone i at the future time t should not be greater than the sum of the available vehicles at node (i, t) and arriving vehicles at node (i, t) . It is important to note that, Constraints (2c) and (2e) are imposed for tracking the vehicle flow into the future to ensure vehicle availability for the accepted pre-booked requests. With the dynamic vehicle swapping strategy for pre-booked requests, the dispatch decision only guarantees that there are available vehicles for the accepted pre-booked requests, but the vehicle-trip assignment is not determined.

Once having made decisions at time τ , the post-decision state of the system can be estimated. To describe the decision at time τ , we define $\mathcal{N}_\tau = \{(i, \tau) | i \in \mathcal{Z}\}$ as the set of nodes at time τ and $\mathcal{E}_\tau = \{e | n_e^+ \in \mathcal{N}_\tau\}$ as the set of arcs where the start nodes are in $\mathcal{N}_\tau$. It is important to emphasize that, the operator only makes dispatch decisions on vehicles available at the current time τ . The dispatch decision $e \in \mathcal{E}_\tau$ specifically for vehicle movements at the current time τ is performed, while vehicle movements at the future time $t > \tau$ are considered to track the vehicle flow to ensure vehicle availability for accepted pre-booked requests. Therefore, the vehicle decision vector at time τ , denoted as x_τ , consists of the individual decision variables $x_{\tau,e}$ for each arc $e \in \mathcal{E}_\tau$. The request decision vector at time τ , denoted as y_τ , consists of the individual decision variables $y_{\tau,i,j,t}$ for each trip request $(i, j, t) \in \mathcal{K}$. The decision vector at time τ is then given by $X_\tau = (x_\tau, y_\tau)$.

The post-decision state, denoted as S_τ^x , represents the state after the decisions have been made. The dispatch decision $e \in \mathcal{E}_\tau$ will change the vehicles' state from n_e^+ to n_e^- , indicating that vehicles become available at node n_e^- after vehicle movements. The transition function for vehicle states can be described as

$$R_{\tau,i,t}^x = R_{\tau,i,t} + \sum_{e \in \mathcal{E}_{i,t}^+ \cap \mathcal{E}_\tau} x_{\tau,e}, \quad \forall (i, t) \in \mathcal{N}, t > \tau \quad (3)$$

It is clear that $R_{\tau,i,\tau}^x = 0$ for any $(i, \tau) \in \mathcal{N}_\tau$.

The transition function for accepted pre-booked requests can be described as

$$D_{\tau,i,j,t}^x = D_{\tau,i,j,t} + y_{\tau,i,j,t}, \quad \forall (i, j, t) \in \mathcal{K}, t > \tau \quad (4)$$

It is clear that $D_{\tau,i,j,\tau}^x = 0$ for any $i, j \in \mathcal{Z}$.

3.3.3. Exogenous information

Exogenous information refers to the latest information received from the environment. The information is used to update the post-decision state S_τ^x to a new state $S_{\tau+1}$. Two types of exogenous information are received and used to update the state of vehicles and trip requests. Let $W_{\tau+1} = (\hat{R}_{\tau+1}, \hat{D}_{\tau+1})$ represent the exogenous information received at time $\tau + 1$.

The exogenous information $\hat{R}_{\tau+1}$ provides an updated estimate of the vehicle state at time $\tau + 1$, which captures the difference in the arrival time of vehicles at their drop-off locations due to factors such as intrazonal travel time. To incorporate this information, $\hat{R}_{\tau+1,i,t}$ represents the changes in the number of vehicles with state (i, t) known at time $\tau + 1$. For example, if a vehicle experiences an arrival delay, its state transitions from (i, t) to $(i, t + 1)$, resulting in $\hat{R}_{\tau+1,i,t} = -1$ and $\hat{R}_{\tau+1,i,t+1} = +1$.

On the other hand, $\hat{D}_{\tau+1,i,j,t}$ refers to the number of new trip requests (i, j, t) received between time τ and $\tau + 1$. This captures the newly arriving requests within the SAV system during this time interval.

3.3.4. Transition function

With the exogenous information, transition functions describe how the state vector evolves from time τ to $\tau + 1$. For the state of vehicles, $R_{\tau+1}$ at time $\tau + 1$ is calculated as

$$R_{\tau+1,i,t} = R_{\tau,i,t}^x + \hat{R}_{\tau+1,i,t}, \quad \forall (i, t) \in \mathcal{N} \quad (5)$$

Regarding the trip requests, the state of accepted trip requests $D_{\tau+1}$ at time $\tau + 1$ is the same as the post-decision state D_τ^x at time τ .

$$D_{\tau+1,i,j,t} = D_{\tau,i,j,t}^x, \quad \forall(i, j, t) \in \mathcal{K} \quad (6)$$

The state variable $\hat{D}_{\tau+1,i,j,t}$ is updated using the exogenous information and captures the number of new trip requests.

3.3.5. Reward function

When a dispatch decision $e \in \mathcal{E}_\tau$ is performed at time τ , it produces revenue or a cost. The net profit c_e represents the reward (i.e., profit) obtained from the dispatch decision e . The reward function for time τ is given by

$$C(S_\tau, X_\tau) = \sum_{e \in \mathcal{E}_\tau} c_e x_{\tau,e} \quad (7)$$

The objective of the problem is to maximize the expected sum of profit across all time points, subjected to the initial state. The objective function can be formulated as

$$\max \mathbb{E} \left\{ \sum_{\tau \in \mathcal{T}} C(S_\tau, X_\tau) | S_0 \right\} \quad (8)$$

This problem is formulated as a dynamic program and divided into multi-stage subproblems at different time points based on Bellman's optimality principle. Each subproblem corresponds to a specific time point, and given a current state S_τ at time τ , the subproblem at time τ can be expressed as a recursion equation.

$$V_\tau(S_\tau) = \max_{X_\tau} (C(S_\tau, X_\tau) + \mathbb{E} \{ V_{\tau+1}(S_{\tau+1}) | S_\tau, X_\tau \}) \quad (9)$$

where $V_{\tau+1}(S_{\tau+1})$ is the value function associated with the state $S_{\tau+1}$. To eliminate the expectation, we can replace the value function of $S_{\tau+1}$ with the value function associated with the post-decision state S_τ^x at time τ .

$$V_\tau(S_\tau) = \max_{X_\tau} (C(S_\tau, X_\tau) + V_\tau^x(S_\tau^x)) \quad (10)$$

$$V_\tau^x(S_\tau^x) = \mathbb{E} \{ V_{\tau+1}(S_{\tau+1}) | S_\tau^x \} \quad (11)$$

where $V_\tau^x(S_\tau^x)$ indicates the expected value being at the state S_τ^x .

4. Approximation dynamic programming

Solving the dynamic programming problem suffers from the curse of dimensionality due to the high-dimensional vectors S_τ and X_τ , making the value function computationally intractable. This section presents an ADP approach by replacing the value function with an approximation for solving the problem.

4.1. Value function approximations

In this subsection, we focus on designing value function approximations (VFA) in the ADP approach. Specifically, we use piecewise-linear functional approximations to effectively capture the spatiotemporal value of vehicles and then develop a mixed integer linear program (MILP) by linearizing the piecewise-linear function.

The system can derive greater value, i.e., higher returns, from an increased number of vehicle resources and trip requests. The structure of the VFA is based on a separable form (Godfrey and Powell, 2002b). The approximation of the value function is represented as $\bar{V}_\tau^x(S_\tau^x)$.

$$\bar{V}_\tau^x(S_\tau^x) = \bar{V}_\tau^x(R_\tau^x) + \bar{V}_\tau^x(D_\tau^x) = \sum_{(i,t) \in \mathcal{N}} \bar{V}_{\tau,i,t}^x(R_{\tau,i,t}^x) + \sum_{(i,j,t) \in \mathcal{K}} \bar{V}_{\tau,i,j,t}^x(D_{\tau,i,j,t}^x) \quad (12)$$

where $\bar{V}_{\tau,i,t}^x(\cdot)$ estimates the expected return produced by the vehicles available at node (i, t) after making decisions at time τ . Similarly, $\bar{V}_{\tau,i,j,t}^x(\cdot)$ estimates the expected return produced by the accepted requests (i, j, t) .

To formulate $\bar{V}_{\tau,i,t}^x(\cdot)$, we use a piecewise-linear concave function because the value function is always piecewise linear and concave in relation to vehicle resources (Godfrey and Powell, 2002a). Let $\bar{v}_{\tau,i,t}^x(q)$ represent the marginal return of the q th vehicle available at node (i, t) after making decisions at time τ . The piecewise-linear value function approximation is expressed by Eq. (13).

$$\bar{V}_{\tau,i,t}^x(R_{\tau,i,t}^x) = \sum_{q=1}^{R_{\tau,i,t}^x} \bar{v}_{\tau,i,t}^x(q) \quad (13)$$

The marginal return of an additional vehicle is influenced by the level of vehicle resources available. As the supply of vehicles increases, the marginal return from adding one more vehicle decreases. Eventually, when the supply reaches a certain level, the marginal return diminishes to zero. This can be expressed as $\bar{v}_{\tau,i,t}^x(q) \geq \bar{v}_{\tau,i,t}^x(q+1)$. For any value q greater than or equal to the upper bound Q on $R_{\tau,i,t}^x$, we have $\bar{v}_{\tau,i,t}^x(q) = 0$.

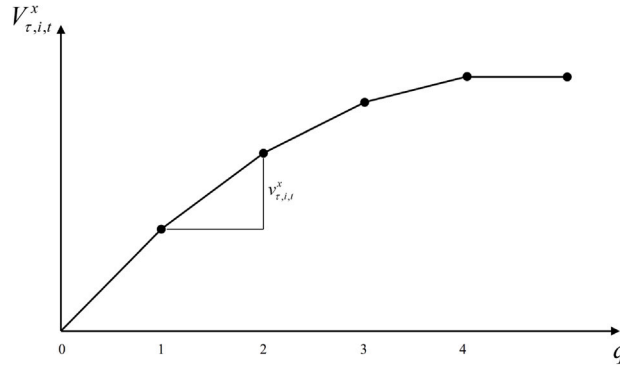


Fig. 2. The piecewise-linear value function.

The piecewise-linear value function is depicted in Fig. 2. In the figure, $\bar{V}_{\tau,i,j,t}^x(0) = 0$, and $\bar{v}_{\tau,i,j,t}^x(q)$ represents the marginal return obtained from adding one more vehicle to the supply of $q - 1$ vehicles, or removing one vehicle from the supply of q vehicles, at node (i, t) .

To formulate $\bar{V}_{\tau,i,j,t}^x(\cdot)$, we use a linear value function since the post-decision state $D_{\tau,i,j,t}^x$, always takes on binary values of 0 or 1. Linear value function approximations are particularly suitable for representing such situations.

$$\bar{V}_{\tau,i,j,t}^x(D_{\tau,i,j,t}^x) = D_{\tau,i,j,t}^x \cdot \bar{v}_{\tau,i,j,t}^x \quad (14)$$

where $\bar{v}_{\tau,i,j,t}^x$ represents the average marginal return of a pre-booked request (i, j, t) .

With the structure of the VFA, the subproblem can be linearized. We introduce a binary variable $\eta_{\tau,i,j,t}^q$, where $q \in \{0, 1, 2, \dots, Q\}$, to indicate the solution of the post-decision state variable $R_{\tau,i,j,t}^x$. The variable $\eta_{\tau,i,j,t}^q$ is equal to 1 if $R_{\tau,i,j,t}^x = q$, and 0 otherwise. We then impose the Constraints (15i) and (15j) to linearize the piecewise-linear concave function. The two constraints together allow for a MILP formulation.

$$\hat{V}_{\tau}(S_{\tau}) = \text{Max} \sum_{e \in \mathcal{E}_{\tau}} c_e x_{\tau,e} + \sum_{(i,j,t) \in \mathcal{K}} \bar{v}_{\tau,i,j,t}^x D_{\tau,i,j,t}^x + \sum_{(i,t) \in \mathcal{N}} \sum_{q=0}^Q \eta_{\tau,i,t}^q \bar{V}_{\tau,i,t}^x(q) \quad (15a)$$

$$\text{s.t. } y_{\tau,i,j,t} \leq \hat{D}_{\tau,i,j,t}, \quad \forall (i, j, t) \in \mathcal{K}, t \geq \tau \quad (15b)$$

$$\sum_{e \in \mathcal{E}_{i,j,\tau}^{S+}} x_{\tau,e} - y_{\tau,i,j,\tau} = 0, \quad \forall (i, j, \tau) \in \mathcal{K} \quad (15c)$$

$$\sum_{e \in \mathcal{E}_{i,j,\tau}^S} x_{\tau,e} = D_{\tau,i,j,\tau}, \quad \forall (i, j, \tau) \in \mathcal{K} \quad (15d)$$

$$\sum_{e \in \mathcal{E}_{i,j,t}^S} x_{\tau,e} - y_{\tau,i,j,t} = D_{\tau,i,j,t}, \quad \forall (i, j, t) \in \mathcal{K}, t > \tau \quad (15e)$$

$$\sum_{e \in \mathcal{E}_{i,\tau}^S} x_{\tau,e} = R_{\tau,i,\tau}, \quad \forall (i, \tau) \in \mathcal{N}_{\tau} \quad (15f)$$

$$\sum_{e \in \mathcal{E}_{i,t}^S} x_{\tau,e} - \sum_{e \in \mathcal{E}_{i,t}^{S+}} x_{\tau,e} \leq R_{\tau,i,t}, \quad \forall (i, t) \in \mathcal{N}, t > \tau \quad (15g)$$

$$R_{\tau,i,t}^x = R_{\tau,i,t} + \sum_{e \in \mathcal{E}_{i,t}^{S+} \cap \mathcal{E}_{\tau}} x_{\tau,e}, \quad \forall (i, t) \in \mathcal{N}, t > \tau \quad (15h)$$

$$D_{\tau,i,j,t}^x = D_{\tau,i,j,t} + y_{\tau,i,j,t}, \quad \forall (i, j, t) \in \mathcal{K}, t > \tau \quad (15i)$$

$$\sum_{q=0}^Q \eta_{\tau,i,t}^q \cdot q = R_{\tau,i,t}^x, \quad \forall (i, t) \in \mathcal{N}, t > \tau \quad (15j)$$

$$\sum_{q=0}^Q \eta_{\tau,i,t}^q = 1, \quad \forall (i, t) \in \mathcal{N}, t > \tau \quad (15k)$$

$$\eta_{\tau,i,t}^q \in \{0, 1\}, \quad \forall (i, t) \in \mathcal{N}, t > \tau \quad (15l)$$

The subproblem makes dispatch decisions for vehicles available at the current time τ . The objective function maximizes the reward generated by the vehicle movements at the current time τ , as well as the expected value of the post-decision state. The two terms ensure that the decisions balance the current reward with the anticipation of future states, contributing to effective decision-making in the real-time vehicle dispatch.

In the subproblem, the parameters related to vehicle supplies and trip requests are all naturally integers. The breakpoints of the piecewise-linear function $\bar{V}_{\tau,i,t}^x(\cdot)$ also occur at integer values. Therefore, the optimal solution of the subproblem is an integer solution. We can relax the integer constraint (15l) according to Proposition 1. This allows us to solve a relaxed version of the subproblem to obtain the dual information.

Proposition 1. *If the breakpoints of the piecewise linear function $\bar{V}_{\tau,i,t}^x(\cdot)$ occur at integer values $R_{\tau,i,t}^x$, then the optimal solution of the subproblem is guaranteed to be an integer.*

Proof. Since $\bar{V}_{\tau,i,t}^x(\cdot)$ is a concave function. We have

$$\sum_{q=0}^Q \eta_{\tau,i,t}^q \cdot \bar{V}_{\tau,i,t}^x(q) \leq \bar{V}_{\tau,i,t}^x\left(\sum_{q=0}^Q \eta_{\tau,i,t}^q \cdot q\right) = \bar{V}_{\tau,i,t}^x(R_{\tau,i,t}^x)$$

Therefore, the optimal solution should guarantee that $\eta_{\tau,i,t}^{q^*} = 1$ where $q^* = R_{\tau,i,t}^x$. $\square$

4.2. DualT and DualNext algorithms

In this subsection, we propose DualT and DualNext algorithms to calibrate the parameters in the VFA.

The general approach of calibrating the VFA parameters involves an iterative process of simulating the system forward in time. During the simulation, a sample provides the exogenous information, and the subproblem (15) is solved for each time $\tau \in \mathcal{T}$. Upon solving the subproblem, dual values are derived from Constraints (15d) to (15g). The dual values represent the marginal return produced by a vehicle or a trip request, and can be used to update the VFA parameters (Simão et al., 2009). The use of the dual values takes advantage of the linear property of the subproblem.

Specifically, at time $\tau + 1$, we solve for the dual value $v_{\tau+1,i,\tau+1}$ from Constraint (15f). The dual value associated with currently available vehicles represents the marginal changes in the objective value $\hat{V}_{\tau+1}$ resulting from a change in $R_{\tau+1,i,\tau+1}$. On the other hand, the parameter $\bar{v}_{\tau,i,\tau+1}^x$ represents the marginal change in $\bar{V}_{\tau}^x$ resulting from a change in $R_{\tau,i,\tau+1}^x$. Notably, $R_{\tau+1,i,\tau+1}$ is considered as a sample of $R_{\tau,i,\tau+1}^x$, and similarly, $\hat{V}_{\tau+1}$ can be regarded as a sample of $\bar{V}_{\tau}^x$. Therefore, for currently available vehicles, the dual value $v_{\tau+1,i,\tau+1}$ derived from Constraint (15f) can be used to update $\bar{v}_{\tau,i,\tau+1}^x$.

$$\bar{v}_{\tau,i,\tau+1}^x = \frac{\partial \bar{V}_{\tau}^x}{\partial R_{\tau,i,\tau+1}^x} \leftarrow \frac{\partial \hat{V}_{\tau+1}}{\partial R_{\tau+1,i,\tau+1}} = v_{\tau+1,i,\tau+1} \quad \forall (i, \tau + 1) \in \mathcal{N}$$

However, for vehicles available at the future time $t > \tau + 1$, the dual value $v_{\tau+1,i,t}$ derived from Constraint (15g) is not suitable for updating $\bar{v}_{\tau,i,t}^x$. The subproblem exclusively makes dispatch decisions for vehicles available at the current time $\tau + 1$, and only the vehicle movements departing at the current time produce rewards in the objective function. Constraint (15g) on the vehicle movements departing at the future time $t > \tau + 1$ are only used to check vehicle availability for accepted pre-booked requests. If there are no pre-booked requests within the system, vehicles available at future times will not receive any dispatch decisions. Consequently, the number of vehicles available at node (i, t) with $t > \tau + 1$ has no impact on the objective function. The dual value $v_{\tau+1,i,t}$ obtained from Constraint (15g) thus does not differ from the previous parameter $\bar{v}_{\tau+1,i,t}^x$.

To address this limitation, we propose two alternative algorithms, i.e., the DualT and DualNext algorithms.

DualT: The DualT algorithm updates the parameter $\bar{v}_{\tau,i,\tau+1}^x$ at time $\tau + 1$ by using the dual value $v_{\tau+1,i,\tau+1}$ derived from Constraint (15f). When the algorithm proceeds to time t , the subproblem is solved to obtain the dual value $v_{t,i,t}$, subsequently updating the parameter $\bar{v}_{t-1,i,t}^x$. In the DualT algorithm, the dual value is also used to update $\bar{v}_{\tau,i,t}^x$. With this algorithm, the parameter value of $\bar{v}_{\tau,i,t}^x$ remains the same across different τ . The intuition behind the DualT algorithm is that the marginal return of a vehicle at node (i, t) can be measured by using the dual information $v_{t,i,t}$ derived from the subproblem at time t . This algorithm extends the “DualMax” algorithm, where the maximal value of $v_{\tau+\mu,i,t}$ for any $\mu = 1, 2, \dots, t - \tau$ is used. The difference is that the “DualT” algorithm only uses the value of $v_{t,i,t}$ (i.e., $\mu = t - \tau$).

DualNext: The DualNext algorithm collects current requests received at time $\tau + 1$ and future requests that will be received at time $\tau + 2$ to $\tau + \bar{t}$, where $\bar{t}$ represents a sufficiently large time exceeding all interzonal travel times. By incorporating future requests, the subproblem makes dispatch decisions for vehicles available at node (i, t) where $t \in \tau + 1, \dots, \tau + \bar{t}$. The dispatch decisions for vehicles available at future times no longer only check vehicle availability for accepted pre-booked requests. Instead, vehicle movements within the time horizon are evaluated in the objective function. Therefore, the subproblem at time $\tau + 1$ maximizes the reward obtained from $\tau + 1$ to $\tau + \bar{t}$, as well as the value function term associated with state S_{τ}^x . For the detailed mathematical formulation of this DualNext algorithm, please refer to the specific formulation in A. In the DualNext algorithm, the dual value $v_{\tau+1,i,t}$ for $t \in \tau + 1, \dots, \tau + \bar{t}$ from Constraint (15g) provides more precise marginal value, and it can be used to update $\bar{v}_{\tau,i,t}^x$ for time $\tau + 1$ to $\tau + \bar{t}$.

Since the VFA of $\bar{V}_{\tau}^x(\cdot)$ is piecewise-linear, the marginal return $\bar{v}_{\tau,i,t}^x$ varies with the number of vehicles q . Let $q_{\tau+1,i,t}$ represent the value of $R_{\tau+1,i,t}$ in this iteration. After solving for the dual value $v_{\tau+1,i,t}$, it indicates the marginal return of one additional vehicle at node (i, t) . We can estimate the marginal value of the $(q_{\tau+1,i,t} + 1)$ -th vehicle using $v_{\tau+1,i,t}$.

To update the parameter, we use a two-sided updating procedure that maintains the concavity of the value functions. If $q > q_{\tau+1,i,t} + 1$, the marginal value of the q th vehicle is 0. Otherwise, if $q \leq q_{\tau+1,i,t} + 1$, the marginal value is $v_{\tau+1,i,t}$.

$$\bar{v}_{\tau,i,t}^x(q) \leftarrow \begin{cases} (1 - \alpha)\bar{v}_{\tau,i,t}^x(q) + \alpha v_{\tau+1,i,t} & \text{if } q \leq q_{\tau+1,i,t} + 1 \\ (1 - \alpha)\bar{v}_{\tau,i,t}^x(q) & \text{if } q > q_{\tau+1,i,t} + 1 \end{cases}$$

where α is a stepsize that dynamically changes with each iteration of the algorithm.

Using a similar approach, we update the parameter $\bar{v}_{\tau,i,j,t}^x$ by utilizing the dual value obtained from Constraint (15d) when solving the subproblem at time t . Consistent with the DualT algorithm, we use the dual value $v_{t,i,j,t}$ to update the parameter $\bar{v}_{\tau,i,j,t}^x$ for any $\tau < t$. Since the VFA of $\bar{V}_{\tau,i,j,t}^x(\cdot)$ is linear, the updating procedure is as follows.

$$\bar{v}_{\tau,i,j,t}^x \leftarrow (1 - \alpha)\bar{v}_{\tau,i,j,t}^x + \alpha v_{t,i,j,t}$$

To summarize, the algorithm for the parameter calibration is outlined in Algorithm 1. The algorithm utilizes a set of training samples Ω to update the parameters in the VFA.

Algorithm 1 The ADP approach

- 1: Initialize VFA parameters: set $\bar{v}_{\tau,i,t}^x$ as 0 and $\bar{v}_{\tau,i,j,t}^x$ as 0.001(a small positive value).
 - 2: **for** each iteration $n = 1 : N$ **do**
 - 3: Initialize the state vector S_0 .
 - 4: **for** each sample $\omega \in \Omega$ **do**
 - 5: **for** $\tau \in \mathcal{T}$ **do**
 - 6: Solve the subproblem in the DualT or DualNext algorithm.
 - 7: Obtain the dual values derived from Constraints (15d) and (15f).
 - 8: Update the state vector using the transition functions (3) to (6).
 - 9: **end for**
 - 10: Update the VFA parameters $\bar{v}_{\tau,i,t}^x$ and $\bar{v}_{\tau,i,j,t}^x$ by using the dual values with the step size α .
 - 11: **end for**
 - 12: Update the step size α .
 - 13: **end for**
-

4.3. ADP approach with lookahead

Once the VFA parameters have been estimated, the ADP approach proceeds to solve the subproblem (15) to determine vehicle dispatch decisions for time τ . The subproblem only considers the current reward obtained at time τ , while the future rewards generated by accepted pre-booked requests are evaluated using the value function. To address this limitation, we propose a lookahead subproblem that incorporates a deterministic lookahead policy. This policy allows for better decision-making by considering future states and rewards.

To incorporate a deterministic lookahead policy, we generate a lookahead horizon $\mathcal{H}_\tau = \{\tau + 1, \dots, \tau + h\}$ at time τ . Here, h represents the size of the lookahead horizon. Unlike the original subproblem (15), which focuses on the dispatch decisions for currently available vehicles, the lookahead subproblem extends its scope to make dispatch decisions over a lookahead horizon, determining vehicle movements from τ to $\tau + h$. The objective function maximizes the reward generated by all these vehicle movements. It directly evaluates the future reward generated by serving the accepted pre-booked requests (i, j, t) for $\tau + 1 \leq t \leq \tau + h$, along with the associated vehicle relocations. Importantly, all these vehicle movements within the lookahead horizon are considered when updating the post-decision state.

To set up the lookahead subproblem, we define $\mathcal{N}_{\tau+h} = \{(i, t) | i \in \mathcal{Z}, \tau \leq t \leq \tau + h\}$ as the set of nodes within the lookahead horizon, and $\mathcal{E}_{\tau+h} = \{e | n_e^+ \in \mathcal{N}_{\tau+h}\}$ as the set of arcs where the start nodes are in $\mathcal{N}_{\tau+h}$, which indicates the vehicle movements within the time horizon. The formulation of the lookahead subproblem is as follows.

$$\text{Max} \quad \sum_{e \in \mathcal{E}_{\tau+h}} c_e x_{\tau,e} + \sum_{(i,j,t) \in \mathcal{K}} \bar{v}_{\tau,i,j,t}^x D_{\tau,i,j,t}^x + \sum_{(i,t) \in \mathcal{N}} \sum_{q=0}^Q \eta_{\tau,i,t}^q \bar{V}_{\tau,i,t}^x(q) \quad (16a)$$

$$\text{s.t. Constraints (15b)–(15g), (15j)–(15k)} \quad (16b)$$

$$R_{\tau,i,t}^x + \sum_{e \in \mathcal{E}_{i,t}^-} x_{\tau,e} = R_{\tau,i,t} + \sum_{e \in \mathcal{E}_{i,t}^+} x_{\tau,e}, \quad \forall (i, t) \in \mathcal{N}, \tau + 1 \leq t \leq \tau + h \quad (16c)$$

$$R_{\tau,i,t}^x = R_{\tau,i,t} + \sum_{e \in \mathcal{E}_{i,t}^+ \cap \mathcal{E}_{\tau+h}} x_{\tau,e}, \quad \forall (i, t) \in \mathcal{N}, t > \tau + h \quad (16d)$$

$$D_{\tau,i,j,t}^x = D_{\tau,i,j,t} + y_{\tau,i,j,t}, \quad \forall (i, j, t) \in \mathcal{K}, t > \tau + h \quad (16e)$$

The objective function (16a) includes the sum of the reward obtained within the time horizon τ to $\tau + h$. Constraints (16c) and (16d) determine the post-decision state $R_{\tau,i,t}^x$, where the dispatching decision for vehicle movements within the time horizon τ to $\tau + h$ affects the post-decision state. Constraint (16c) specifies that for any time t from $\tau + 1$ to $\tau + h$, the number of available vehicles

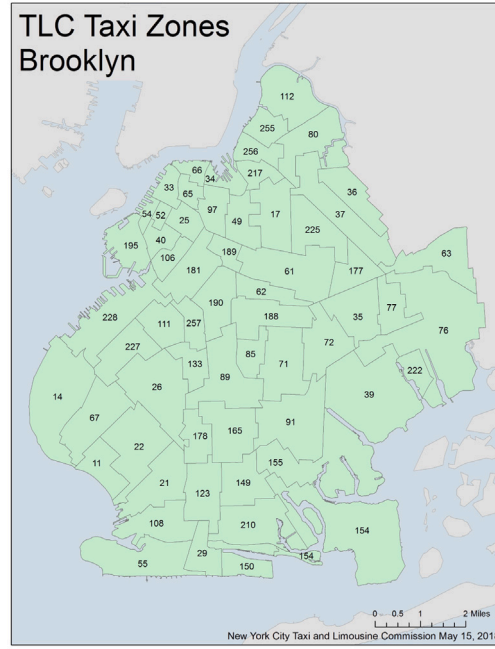


Fig. 3. Brooklyn partitioned into taxi zones.

at node (i, t) after the decision is determined by the initially available vehicles at node (i, t) , the arriving vehicles at node (i, t) , and the departing vehicles from node (i, t) . After the decision, there still are vehicles available at node (i, t) for $\tau + 1 \leq t \leq \tau + h$. Constraint (16d) indicates that for any time t after time $\tau + h$, the number of available vehicles at node (i, t) after the decision is influenced by the vehicles departing during the time interval τ to $\tau + h$. Constraint (16e) updates the post-decision state of accepted pre-booked requests (i, j, t) for $t > \tau + h$. The difference from the subproblem in the DualNext algorithm is that the lookahead subproblem has no future request information.

After solving the lookahead subproblem, we update the post-decision state by only using the dispatch decisions $e \in \mathcal{E}_\tau$. The dispatch decisions in the lookahead horizon $\mathcal{H}_\tau$ will not be performed.

5. Numerical study

To demonstrate the ADP approach, we conduct a numerical study based on real taxi data in Brooklyn, New York.

5.1. Numerical setting

As shown in Fig. 3, the geographic region of Brooklyn is partitioned into 61 taxi zones by the New York City Taxi & Limousine Commission (TLC). We assume the interzonal travel distance between two zones is estimated using 1.2 times the Euclidean distance of the zone centroids and the average driving speed of vehicles is 20 km/h. The study period of interest is from 6:00 to 12:00 and it is divided into 72 time intervals with 5 min per time interval.

To generate trip requests for our experiments, we utilize a green taxi trip dataset provided by the TLC. The dataset consists of trip records specifically from Tuesdays, Wednesdays, and Thursdays between July and September 2019. We focus on trips originating and terminating in Brooklyn within this study period. Each trip record contains various details including pick-up and drop-off times, pick-up and drop-off zones, trip distances, and other relevant information. Each trip is converted into a trip request, which includes information on the departure time, pick-up zone, and drop-off zone. The departure time is rounded to align with discrete time intervals. The arrival time of vehicles at the drop-off locations is estimated from the interzonal travel time. We generate a total of 30 samples, taking different days into account. Among these samples, 20 are allocated for training purposes, allowing us to approximate the value function. The remaining 10 samples are reserved for testing, enabling us to evaluate the effectiveness of the proposed ADP approach. The spatial distribution of pick-up and drop-off demand can be observed in Fig. 4. The average number of requests in the testing samples is plotted in Fig. 5. The average trip distance is calculated to be 5.10 km.

The fleet size of vehicles is set to 120. The initial locations are determined according to the distribution of pick-up locations. Vehicles receive revenue of \$1.5/km when serving a request from users and incur a travel cost of \$0.6/km when moving for service and relocation, mainly including depreciation, maintenance, and fuel costs. The net profit of vehicles for service and relocation is \$0.9/km and \$-0.6/km, respectively.

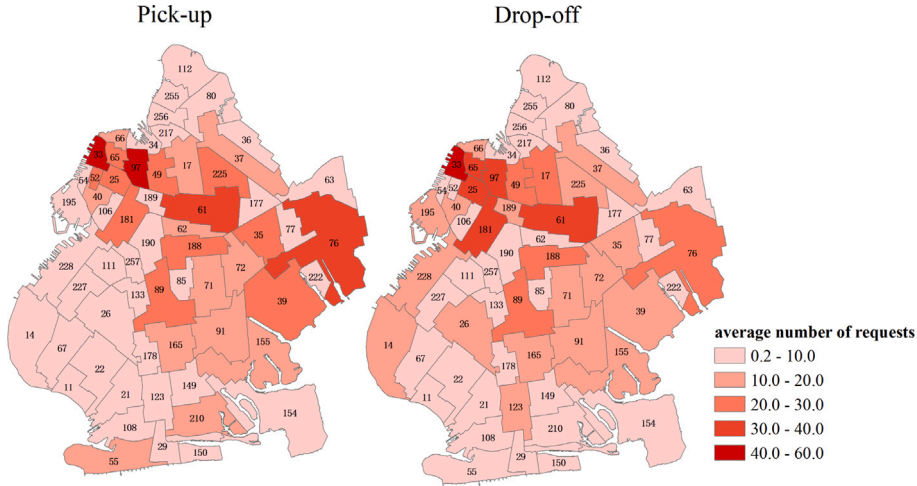


Fig. 4. Spatial distribution of pick-up and drop-off demand.

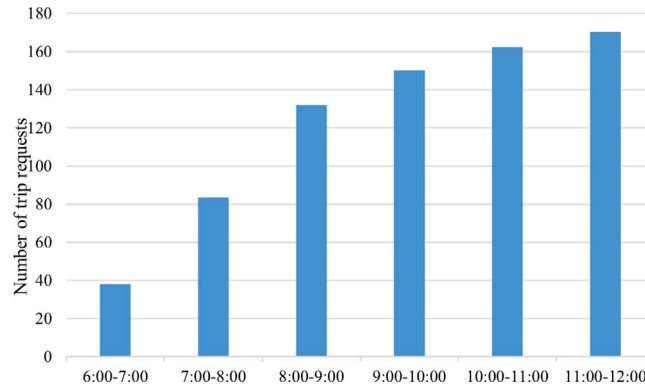


Fig. 5. Average number of requests per hour.

We examine three problem instances representing various reservation modes. The first instance, labeled as the “0% instance”, exclusively consists of on-demand trips, meaning there are no pre-booked requests. The other two instances include reservations, with the percentage of pre-booked requests set at approximately 20% and 40%, respectively. For on-demand requests, we assume that the maximum allowable waiting time for users is 5 min. Regarding pre-booked requests, a valid request time is considered to be at least 30 min ahead of the desired departure time. Consequently, we randomly select 20% or 40% of requests as pre-booked requests, with the request time randomly generated between 6:00 and 30 min before the desired departure time.

We execute the proposed ADP approach on a personal computer with an Intel Xeon E5-2680 2.5G CPU and 16G memory, in which the ILOG CPLEX 12.7 is applied as an LP solver.

5.2. Parameter calibration

The DualT algorithm and the DualNext algorithm are first performed to calibrate the VFA parameters. The number of iterations is set to 20 and 10 in the two training algorithms. Throughout the iterations, each training sample provides trip request information, and the subproblems at each time point are solved to obtain the dual values. Once 5 training samples are computed, the average dual values are used to update the VFA parameters. At the first iteration, the step size is initially set at 0.3 and it decreases by 5% during each subsequent iteration. In the DualNext algorithm, the preset value of $\bar{t}$ is set to 12, i.e., one hour. To prevent overfitting in the training algorithms, the dual values obtained by a single sample are not directly used to update VFA parameters. Instead, the average dual values from five training samples are used. Additionally, an adaptive step size is employed, which is set to be decreased with each iteration.

The training curves of the DualT and DualNext algorithms are illustrated in Fig. 6. In the DualT algorithm, the average total reward of the training samples demonstrates improvement during the training iterations. After approximately 15 iterations, the average total reward stabilizes without any decrease, indicating successful training to calibrate the VFA parameters. On the other hand, in the DualNext algorithm, the average reward shows a slight increase in the first two iterations and then almost remains the

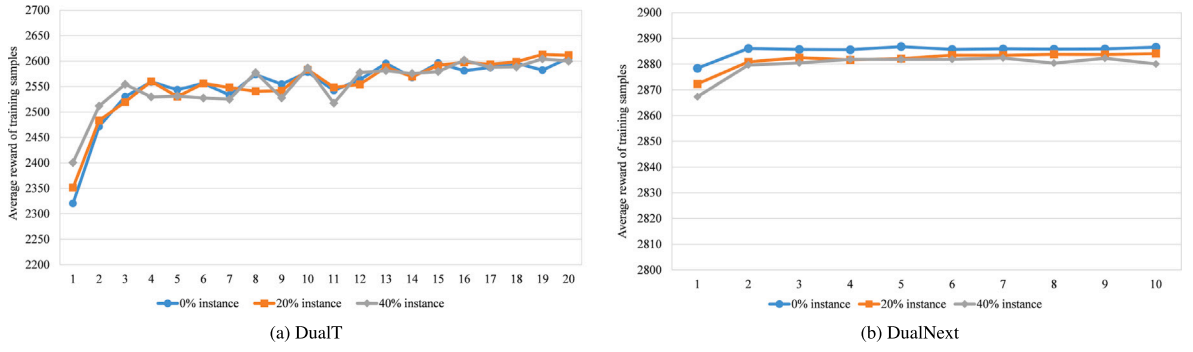
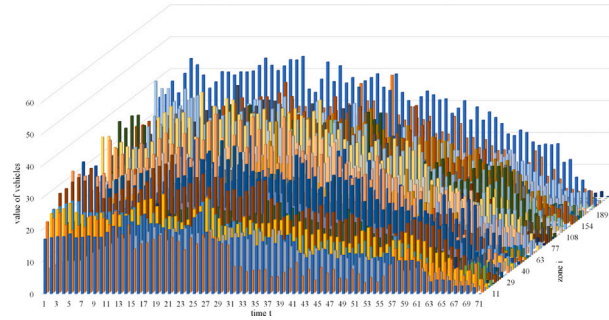
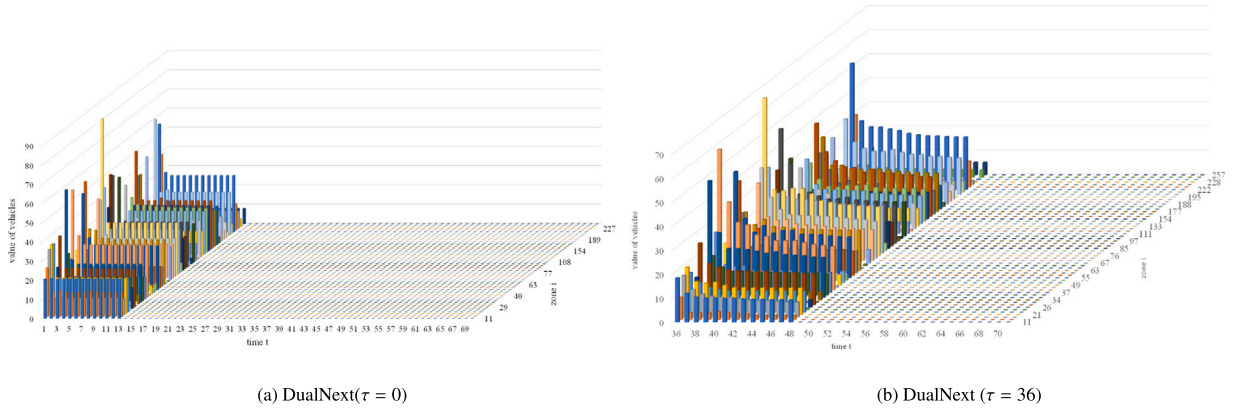


Fig. 6. Training curves of the DualT and DualNext algorithms.

Fig. 7. VFA parameters $\bar{v}_{\tau,i,t}^x$ obtained by the DualT algorithm.Fig. 8. VFA parameters $\bar{v}_{\tau,i,t}^x$ obtained by the DualNext algorithm.

same. This is because the DualNext algorithm utilizes exact future request information when solving the current decision, allowing it to find near-optimal solutions.

After training, we obtain the value of the VFA parameters. The parameter $\bar{v}_{\tau,i,t}^x$ represents the marginal return of a vehicle available at node (i, t) after decision at time τ . The parameter values obtained by the DualT and DualNext algorithms in the 0% instance are depicted in Figs. 7 and 8, respectively. The parameter values at time $\tau = 0$ (i.e., 6:00) and $\tau = 36$ (i.e., 9:00) are compared.

The parameter value $\bar{v}_{\tau,i,t}^x$ of node (i, t) obtained by the DualT algorithm remains constant, regardless of time τ at which the decision is made. Conversely, the parameter value obtained by the DualNext algorithm is dependent on time τ . At $\tau = 0$, the marginal return for vehicles available at times $t > 13$ is set to zero. This is because the DualNext algorithm collects the information of trip requests departing within the horizon from time $t = 1$ to 12. As the time proceeds to $\tau = 36$, the marginal return for the vehicles is changed to accommodate the evolving of the system.

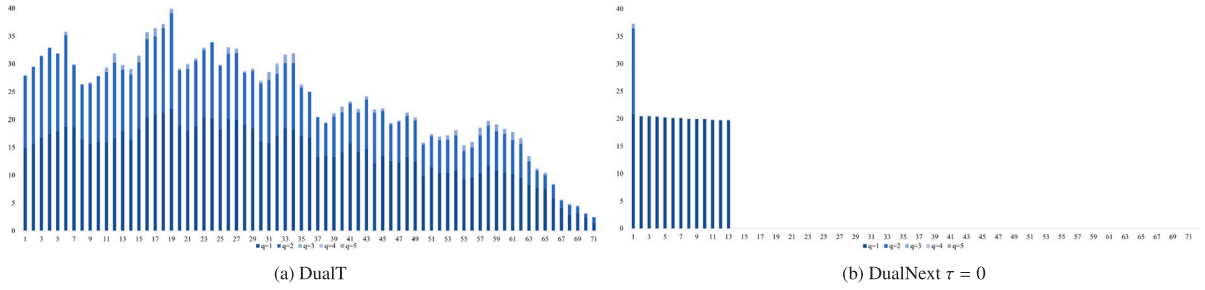
Fig. 9. VFA parameters $\bar{p}_{\tau,t,d}^x$ for Zone 33.

Table 1
Comparisons of different ADP approaches.

		DualT		DualNext	
		ADP	ADP with lookahead	ADP	ADP with lookahead
0% instance	Reward	2639.3	2516.2	2618.5	2664.9
	Fulfillment rate	94.11%	86.01%	93.07%	91.82%
	Relocation trips	229.0	89.7	276.0	187.8
20% instance	Reward	2642.8	2598.2	2620.7	2729.9
	Fulfillment rate	94.43%	89.49%	94.32%	95.38%
	Relocation trips	287.6	169.3	346.8	266.2
40% instance	Reward	2653.9	2668.1	2609.0	2757.9
	Fulfillment rate	95.23%	92.18%	92.99%	96.51%
	Relocation trips	326.0	229.7	350.5	311.7

For clarity, we take Zone 33 as an illustrative example, with the parameter values depicted in Fig. 9. In the DualT algorithm, the parameter values are primarily influenced by trip requests departing from zone i at time t . More trip requests departing at a specific time t can lead to a larger return, as vehicles have more opportunities to serve requests and earn higher rewards. Moreover, the parameter value tends to decrease as time t increases within a zone. This is because vehicles available at earlier times can serve more requests, resulting in a higher return. However, it does not strictly follow the monotonicity property, where the parameter value decreases with time t increasing. In contrast, in the DualNext algorithm, the parameter value at one zone i strictly follows a descending trend as time t increases. This is consistent with the monotonicity property in the study (Jiang and Powell, 2015).

Comparing the two training algorithms, the DualNext algorithm tends to exhibit a higher marginal return than the DualT algorithm. This can be attributed to the fact that the DualNext algorithm at time $\tau+1$ collects upcoming information on trip requests that will be received within the horizon from time $\tau+2$ to $\tau+\bar{t}$. The marginal return is directly estimated according to the future requests. In contrast, the DualT algorithm only collects the trip requests received at the current time t , without accounting for future requests. Consequently, the DualNext algorithm places a greater emphasis on maximizing return compared to the DualT algorithm.

5.3. Performance of the ADP approach

With the VFA parameters, the ADP approach uses the original subproblem for vehicle dispatch decisions across the 10 testing samples. To incorporate the lookahead strategy within the ADP approach, the lookahead subproblem is used to make vehicle dispatch decisions. The lookahead horizon is set to 30 min. The objective value (i.e., the average reward) obtained by the ADP approaches is taken to assess their performance. The fulfillment rate (i.e., the percentage of fulfilled trip requests) and the number of relocation trips are measured as well. Table 1 compares the performance of four ADP approaches: (i) the ADP approach using DualT, (ii) the ADP approach using DualNext, (iii) the ADP approach with lookahead using DualT, and (iv) the ADP approach with lookahead using DualNext.

We compare the performance of two ADP approaches without lookahead. The ADP approaches using the two training algorithms obtain nearly the same objective value of the average rewards. The fulfillment rate reaches about 93% to 95% across different instances, with the DualT algorithm outperforming the DualNext algorithm by roughly 1%. The DualT algorithm executes fewer relocation trips due to the smaller value of vehicles' marginal return, thereby saving on relocation costs.

However, upon incorporating the lookahead strategy, the ADP approach using DualT does not exhibit better performance in the 0% and 20% instances. The average reward declines by 4.7% and 1.9%, respectively. This decline is attributed to the lookahead subproblem, which makes dispatch decisions over a lookahead horizon and directly includes associated relocation costs in the objective function. Consequently, there are significantly fewer relocations, leading to lower fulfillment rates. In contrast, in the 40% instance, the ADP approach with lookahead can achieve 0.5% higher rewards than the ADP approach without lookahead. The lookahead strategy is effective when dealing with a significant proportion of pre-booked requests.

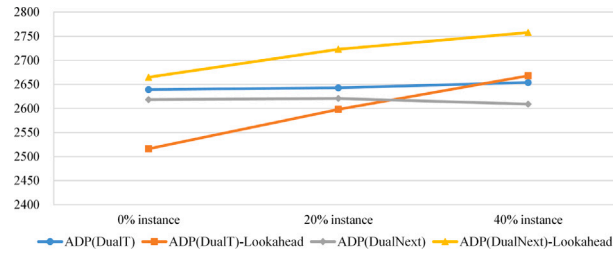


Fig. 10. Operational profit in the three instances.

Table 2
The impact of different sizes of the lookahead horizon.

Lookahead horizon		30 min	45 min	60 min
0% instance	Reward	2664.9	2662.1	2658.3
	Fulfillment rate	91.82%	91.73%	91.60%
	Relocation trips	187.8	186.1	184.6
20% instance	Reward	2729.9	2732.5	2732.5
	Fulfillment rate	95.38%	95.62%	95.54%
	Relocation trips	266.2	270.7	273.3
40% instance	Reward	2757.9	2753.6	2759.1
	Fulfillment rate	96.51%	96.38%	96.61%
	Relocation trips	311.7	315.1	314.1

On the other hand, when using the DualNext algorithm, the ADP approach with lookahead consistently performs better than the ADP approach without lookahead in all instances. The ADP approach with lookahead significantly improves the average reward by 1.8%, 4.2%, and 5.7%, respectively. With a higher percentage of pre-booked requests, the improvement is more significant. In the DualNext algorithm, the number of relocation trips is also decreased when incorporating the lookahead strategy. However, the decrease is less significant in the DualNext algorithm than in the DualT algorithm. The superior performance of the DualNext algorithm can be attributed to the monotonicity property of the VFA parameters.

In summary, the ADP approach with lookahead, using the DualNext algorithm, demonstrates the most effective performance. This demonstrates the effectiveness of incorporating the lookahead strategy to enhance the overall performance of the ADP approach. Particularly in the 40% instance, the ADP approach with lookahead consistently achieves higher rewards compared to the ADP approach without lookahead, regardless of whether the DualT or DualNext algorithm is used. For a SAV system incorporating a percentage of pre-booked requests, a lookahead strategy is recommended for decision-making.

The operational profit obtained in the three instances is shown in Fig. 10. Despite the identical trip demand in the three instances, it is evident that in the 40% instance, the SAV system achieves the highest profit, particularly incorporating the lookahead strategy. This is because the SAV system collects information about future trip demand departing at future times through reservations, and then utilizes the future trip demand to optimize the current vehicle dispatch. As a result, there is a notable advantage in encouraging users to pre-book their trips. By promoting trip reservations, the system enhances the fulfillment rate, increases the operational profit, and reduces user waiting times. The trip reservation proves beneficial for both users and operators.

To further investigate the lookahead strategy in the ADP approach, we conduct experiments with different sizes of the lookahead horizon. Specifically, the lookahead horizon is varied, with settings of 30, 45, and 60 min, respectively. In these experiments, the DualNext algorithm is used as the training algorithm. The results are presented in Table 2.

When using different sizes of lookahead horizons, the ADP approach yields a very slight difference in the operational results. The performance of the ADP approach remains stable when the lookahead horizon exceeds 30 min. This is because, although the subproblem makes dispatch decisions over the lookahead horizon, only the dispatch decisions on vehicle movements at the current time are executed. It demonstrates that the lookahead horizon is effective when set at 30 min.

5.4. Comparison to benchmark policies

To further examine the effectiveness of the ADP policy, we implement four policies as benchmarks.

- The first one is a myopic policy that only maximizes the current reward. The myopic policy prioritizes serving current trip requests and attempts to hold vehicle inventory rather than executing relocations. To make responses to the pre-booked trip requests, the myopic policy will check whether there are available vehicles to serve them, and the pre-booked requests will be accepted if there are. The myopic policy is represented by the ADP approach using the initialized VFA parameters, without training.

Table 3
Comparisons with benchmark policies.

Policies		Myopic	DL	ADP	ADP-lookahead	SP	PI
0% instance	Reward	2190.2	2190.2	2639.3	2664.9	2667.7	2887.1
	Fulfillment rate	74.12%	74.12%	94.11%	91.82%	92.17%	98.85%
	Relocation trips	0	0	229.0	187.8	223.5	334.6
20% instance	Reward	2254.2	2343.6	2642.8	2729.9	2734.2	2890.7
	Fulfillment rate	77.55%	80.57%	94.43%	95.38%	94.77%	99.02%
	Relocation trips	108.7	128.8	287.6	266.2	274.3	347.8
40% instance	Reward	2294.7	2536.9	2653.9	2757.9	2761.8	2894.2
	Fulfillment rate	80.47%	88.92%	95.23%	96.51%	95.42%	99.39%
	Relocation trips	179.5	278.1	326.0	311.7	292.6	353.5

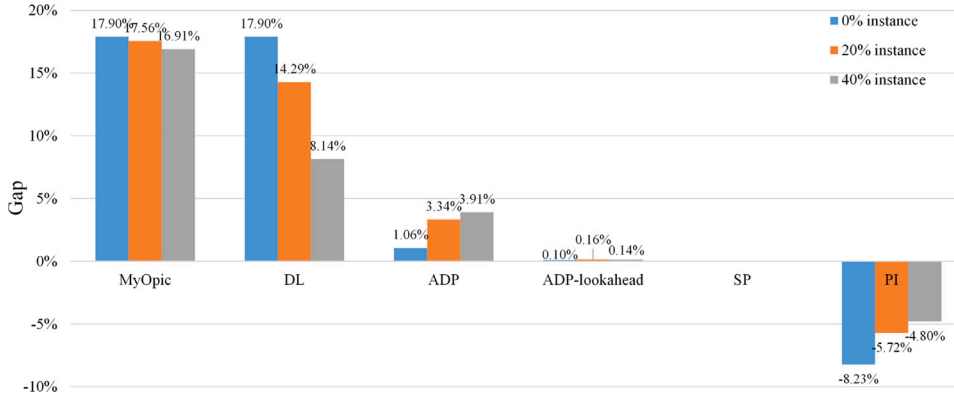


Fig. 11. Gaps in the average reward.

- The second policy is a deterministic lookahead (DL) policy. The DL policy optimizes the dispatch decisions from the current time to the future with the objective of maximizing the sum of rewards over all the time points. The DL policy will consider the accepted pre-booked requests departing at future times but does not consider trip requests received at future times. The DL policy is represented by the ADP approach with lookahead, also using the initialized VFA parameters.
- The third policy is a stochastic programming (SP) policy with lookahead. The SP policy also optimizes the dispatch decisions from the current to the future. However, it incorporates historical samples to represent realizations of future received requests. Considering the computation burdens, the SP policy at time τ involves the future requests received within the lookahead horizon H_τ . Requests received after this horizon are not taken into account. A two-stage stochastic programming formulation is developed in this policy.
- Lastly, we consider an optimal policy under perfect information (PI). The PI policy assumes that the system has complete information on all trip requests, including origins, destinations, and departure times. The PI policy uses the perfect information to make the optimal decisions.

The mathematical formulation of the decision-making subproblems for time τ is presented in B. The myopic and DL policies are two baseline policies that do not consider historical data, thereby providing lower bounds. The PI policy sets a theoretical upper bound by considering perfect request information to make the optimal vehicle dispatch decisions. However, obtaining precise future request information is impractical in practice. To provide an estimate of the upper bound, the SP policy employs 10 training samples to simulate future request information and calculates the sample average.

To conduct a comparison, the ADP approach using the original subproblem and the ADP approach with the lookahead strategy is compared to the benchmark policies. The average results of the 10 testing samples are listed in Table 3.

Taking the reward result obtained by the SP policy as the upper bound, the gaps of the other policies are computed, which is indicated in Fig. 11.

The myopic and DL policies provide lower bounds for the average reward. In the 0% instance, where there are no pre-booked requests, both policies essentially yield the same results. The myopic policy prioritizes maximizing the current reward without executing any relocation trips, resulting in a substantial gap of 17.9%. In the 20% and 40% instances, the myopic policy exhibits a similar gap, reaching about 17%. However, the DL policy achieves improvement with higher rewards and fulfillment rates, reducing the gap to 14.29% and 8.14%, respectively. This improvement becomes more pronounced with a higher percentage of pre-booked requests, emphasizing the effectiveness of incorporating the lookahead strategy.

In comparison to the myopic policy, the ADP approach based on the VFA consistently demonstrates superior decision-making. The gaps in the three instances are significantly reduced from about 17% to less than 4%. The value function in the ADP approach

Table 4
Computation time for each decision (s).

Policies	Myopic	DL	ADP	ADP-lookahead	SP	PI
0% instance	0.53	0.22	0.55	0.22	5.7	0.95
20% instance	0.56	0.23	0.57	0.24	5.2	0.82
40% instance	0.57	0.22	0.71	0.24	6.8	0.93

enables current vehicle dispatch to anticipate the future, leading to improved decisions. Moreover, the ADP approach with lookahead outperforms the DL policy, further emphasizing the better performance of the ADP approach. Due to the introduction of the VFA, the decision-maker balances the current rewards with the anticipation of future dispatch, contributing to effective decision-making in the real-time dispatch process.

Compared to the SP policy, the ADP approach with lookahead obtains the average reward with a gap of less than 0.2%. The ADP approach with lookahead can achieve the operational results comparable to the SP policy, highlighting its high performance.

In terms of computational efficiency, Table 4 lists the average computation time for each decision. The computation time for different instances remains nearly the same. All the decisions take less than 1 s, except that the SP policy takes approximately 5 to 7 s. The ADP approach with lookahead requires less than 0.3 s for each decision, significantly faster than the SP policy. Therefore, the ADP approach with lookahead achieves comparable results to the SP policy but with significantly lower computation time, which proves to be well-suited for the real-time vehicle dispatch problem.

6. Conclusions

This paper presents an ADP approach to solve a real-time dispatch problem for SAVs under the hybrid request behavior with mixed on-demand and pre-booked requests. The problem is formulated using a dynamic program framework, where a subproblem model for each time stage is formulated. To handle the high-dimensional state space, piecewise-linear and linear functional approximations are constructed for the value function. The DualT and DualNext algorithms are proposed to update the value function parameters. Additionally, a lookahead strategy is integrated into the subproblems to enhance the ADP approach.

To evaluate the performance of the ADP approach, numerical experiments are conducted based on taxi data in Brooklyn. Benchmark policies including myopic, deterministic lookahead, stochastic lookahead, and optimal policies are designed to compare against the ADP approach. Results show that the ADP approach outperforms the myopic policies. Furthermore, when the lookahead strategy is incorporated, the ADP approach achieves comparable results to the stochastic programming policy but with significantly lower computation time. The numerical results highlight the high performance of the ADP approach.

From a managerial perspective, this paper emphasizes the importance of considering the anticipation of the future in current vehicle dispatch. If a SAV system incorporates a percentage of pre-booked requests, a lookahead strategy is recommended for decision-making, consistently achieving both better operational rewards and fulfillment rates. Moreover, a higher percentage of pre-booked requests results in better operational results. Encouraging users to pre-book their trips can lead to improved system performance, benefiting both users and operators.

This paper studies the real-time dispatch problem for SAVs. It can be extended by considering electric SAVs, in which the charging decisions should be made. Besides, the potential impact of SAV fleets offering ridesharing services could be explored in future research.

CRedit authorship contribution statement

Yao Chen: Conceptualization, Formal analysis, Investigation, Methodology, Software Validation, Visualization, Writing – original draft. **Yang Liu:** Conceptualization, Funding acquisition, Methodology, Validation, Writing – review & editing. **Yun Bai:** Validation, Writing – review & editing. **Baohua Mao:** Resources, Writing – review & editing.

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Appendix A. Subproblem in the DualNext algorithm

In the DualNext algorithm, we assume that at time τ the system collects future request information from time $\tau + 1$ to $\tau + \bar{t}$. By incorporating future requests, the subproblem makes dispatch decisions for vehicles available at node (i, t) where $t \in \tau + 1, \dots, \tau + \bar{t}$. The reward generated by the vehicle movements within the time horizon from $\tau + 1$ to $\tau + \bar{t}$ is considered in the objective function. These vehicle movements also affect the post-decision state S_{τ}^x .

$$\text{Max} \sum_{e \in \mathcal{E}_{\tau+\bar{t}}} c_e x_{\tau,e} + \sum_{(i,j,t) \in \mathcal{K}} \bar{v}_{\tau,i,j,t}^x D_{\tau+i,j,t}^x + \sum_{(i,t) \in \mathcal{N}} \sum_{q=0}^Q \eta_{\tau,i,t}^q \bar{v}_{\tau+i,t}^x(q) \quad (\text{A.1})$$

s.t. Constraints(15b)–(15g), (15j)–(15k) (A.2)

$$\sum_{e \in \mathcal{E}_{i,j,t}^{S+}} x_{\tau,e} \leq \sum_{\tau+1 \leq t' \leq \tau+\bar{t}} D_{\tau',i,j,t}, \quad \forall (i,j,t) \in \mathcal{K}, \tau+1 \leq t \leq \tau+\bar{t} \quad (\text{A.3})$$

$$\sum_{e \in \mathcal{E}_{i,t}^{S-}} x_{\tau,e} - \sum_{e \in \mathcal{E}_{i,t}^{S+}} x_{\tau,e} = R_{\tau,i,t}, \quad \forall (i,t) \in \mathcal{N}, \tau+1 \leq t \leq \tau+\bar{t} \quad (\text{A.4})$$

$$R_{\tau,i,t}^x + \sum_{e \in \mathcal{E}_{i,t}^{S-}} x_{\tau,e} = R_{\tau,i,t} + \sum_{e \in \mathcal{E}_{i,t}^{S+}} x_{\tau,e}, \quad \forall (i,t) \in \mathcal{N}, \tau+1 \leq t \leq \tau+\bar{t} \quad (\text{A.5})$$

$$R_{\tau,i,t}^x = R_{\tau,i,t} + \sum_{e \in \mathcal{E}_{i,t}^{S+} \cap \mathcal{E}_{\tau+\bar{t}}} x_{\tau,e}, \quad \forall (i,t) \in \mathcal{N}, t > \tau+\bar{t} \quad (\text{A.6})$$

$$D_{\tau,i,j,t}^x = D_{\tau,i,j,t} + y_{\tau,i,j,t}, \quad \forall (i,j,t) \in \mathcal{K}, t > \tau+\bar{t} \quad (\text{A.7})$$

The difference from the lookahead subproblem is that this subproblem includes additional Constraints (A.1c) and (A.1d). Constraint (A.1c) ensures that the vehicle flow in service arcs cannot exceed the number of requests (i,j,t) for $\tau+1 \leq t \leq \tau+h$. Constraint (A.1d) is the flow conservation constraint to ensure that all the vehicles available within the time horizon from $\tau+1$ to $\tau+\bar{t}$ are dispatched. After the decision, there are no vehicles available at node (i,t) for $\tau+1 \leq t \leq \tau+\bar{t}$.

Appendix B. Benchmark policies

B.1. Myopic subproblem

The myopic subproblem at time τ is formulated as Model (B.1). The parameter ϵ is a small positive value to ensure that the received pre-booked requests will be accepted if there are available vehicles.

$$\text{Max} \sum_{e \in \mathcal{E}_{\tau}} c_e x_{\tau,e} + \epsilon \sum_{(i,j,t) \in \mathcal{K}} y_{\tau,i,j,t} \quad (\text{B.1a})$$

$$\text{s.t. Constraints (15b)–(15f),} \quad (\text{B.1b})$$

B.2. DP subproblem

The DP subproblem at time τ is formulated as Model (B.2). The objective is to maximize the sum of rewards over all the time points from time τ .

$$\text{Max} \sum_{\tau \in \mathcal{T}} \sum_{e \in \mathcal{E}_{\tau}} c_e x_{\tau,e} \quad (\text{B.2a})$$

$$\text{s.t. Constraints (15b)–(15f),} \quad (\text{B.2b})$$

B.3. SP subproblem

The SP subproblem at time τ is formulated as Model (B.3). The model is a two-stage program, in which the decision variable $x_{\tau,e}^{\omega}$ is introduced to indicate the arc flow in sample ω . For each sample ω , it should satisfy Constraints (15b)–(15f). Constraint (B.3h) is the nonanticipativity constraint to guarantee that the first-stage variables are identical across the different samples.

$$\text{Max} \sum_{\omega} \sum_{\tau \in \mathcal{T}} \sum_{e \in \mathcal{E}_{\tau}} c_e x_{\tau,e}^{\omega} \quad (\text{B.3a})$$

$$\text{s.t. } y_{\tau,i,j,t} \leq \hat{D}_{\tau,i,j,t}, \quad \forall (i,j,t) \in \mathcal{K}, t \geq \tau \quad (\text{B.3b})$$

$$\sum_{e \in \mathcal{E}_{i,j,\tau}^{S+}} x_{\tau,e}^{\omega} - y_{\tau,i,j,\tau} = 0, \quad \forall (i,j,\tau) \in \mathcal{K} \quad (\text{B.3c})$$

$$\sum_{e \in \mathcal{E}_{i,j,\tau}^{S-}} x_{\tau,e}^{\omega} = D_{\tau,i,j,\tau}, \quad \forall (i,j,\tau) \in \mathcal{K} \quad (\text{B.3d})$$

$$\sum_{e \in \mathcal{E}_{i,j,t}^{S-}} x_{\tau,e}^{\omega} - y_{\tau,i,j,t} = D_{\tau,i,j,t}, \quad \forall (i,j,t) \in \mathcal{K}, t > \tau \quad (\text{B.3e})$$

$$\sum_{e \in \mathcal{E}_{i,t}^{S-}} x_{\tau,e}^{\omega} - \sum_{e \in \mathcal{E}_{i,t}^{S+}} x_{\tau,e}^{\omega} = R_{\tau,i,t}, \quad \forall (i,t) \in \mathcal{N}, t \geq \tau \quad (\text{B.3f})$$

$$\sum_{e \in \mathcal{E}_{i,j,t}^{S+}} x_{\tau,e}^{\omega} \leq \sum_{\tau+1 \leq t' \leq \tau+h} D_{\tau',i,j,t}, \quad \forall (i,j,t) \in \mathcal{K}, \tau+1 \leq t \leq \tau+h \quad (\text{B.3g})$$

$$x_{\tau,e}^{\omega} = x_{\tau,e}, \quad \forall e \in \mathcal{E}_{\tau} \quad (\text{B.3h})$$

B.4. PI subproblem

The PI subproblem at time τ is formulated as Model (B.4). Considering the computation burdens, the PI policy at time τ involves the exact request information received in the horizon from time τ to $\tau + h$.

$$\hat{V}_{\tau}(S_{\tau}) = \text{Max} \sum_{e \in \mathcal{E}_{\tau+h}} c_e x_{\tau,e} + \epsilon \sum_{(i,j,t) \in \mathcal{K}} y_{\tau,i,j,t} \quad (\text{B.4a})$$

$$\text{s.t. Constraints (15b)–(15f),} \quad (\text{B.4b})$$

$$\sum_{e \in \mathcal{E}_{i,j,t}^{S+}} x_{\tau,e} \leq \sum_{\tau+1 \leq \tau' \leq \tau+h} D_{\tau',i,j,t}, \forall (i,j,t) \in \mathcal{K}, \tau+1 \leq t \leq \tau+h \quad (\text{B.4c})$$

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